

# FANGTONG ZHOU

2040 Torgersen Hall, Blacksburg, VA 24061 USA

Email: fangtongz@vt.edu, Phone: +1 540-824-9938, Website: <https://frontop137.github.io>

## EDUCATION

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**Virginia Tech**, Blacksburg, VA, USA

2024.8 - present

- Ph.D. in Computer Engineering
- GPA: 4.0/4.0
- Advisor: Prof. Tom Hou

**ShanghaiTech University**, Shanghai, China

2021.9 - 2024.6

- M.S. in Information Engineering
- GPA: 3.78/4.0
- Advisor: Prof. Yang Yang, Prof. Yong Zhou

**South China University of Technology**, Guangzhou, China

2017.9 - 2021.6

- B.Eng. in Information Engineering
- GPA: 3.75/4.0 (89.00/100)
- Ranking: 8/233

## RESEARCH EXPERIENCE

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- **AirComp-assisted Hierarchical Personalized Federated Learning [1][4]** 2022.6 - 2024.6
  - Proposed an AirComp-assisted hierarchical personalized FL that simultaneously learns global and personalized models while mitigating interference through cloud/edge beamforming design, significantly improving convergence and accuracy in heterogeneous wireless networks
- **Edge Interval Control in Hierarchical Federated Learning [2]** 2023.3 - 2023.6
  - Developed an AirComp-assisted hierarchical FL that jointly optimizes edge aggregation intervals and device transceiver design via relaxation-rounding and Lyapunov-based algorithms, achieving faster convergence and higher accuracy under wireless communication constraints
- **Decentralized Satellite Federated Learning [5][8]** 2023.6 - 2024.6
  - Introduced a multi-orbit decentralized satellite FL framework that leverages intra- and inter-orbit ISLs for model aggregation without ground stations, analyzed convergence to guide local iteration settings, and developed a JRARS algorithm for joint routing, bandwidth, and power optimization, achieving faster convergence and lower energy consumption in LEO constellations
- **WOS+ [6][9]** 2025.1 - 2025.6
  - Proposed Wait-for-Optimal-Set (WOS+), a semi-asynchronous FL scheme that adaptively selects clients based on computation latency and model-version gap while employing dynamic resource block allocation, achieving faster convergence and higher accuracy than existing methods in dynamic wireless networks
- **FedHusky [7]** 2025.6 - 2025.9
  - Proposed FedHusky, an FL system that employs a calendar-based client scheduling, optimization-driven group formation, and dynamic birth-death processes to maximize client utilization, significantly improving convergence speed and training efficiency under small and heterogeneous datasets

- Proposed FedLAWN, a split FL algorithm tailored to low-altitude wireless networks, in which aerial clients have transient connections with the BS and limited computation resources.

## PUBLICATIONS

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- [1] F. Zhou, Z. Wang, X. Luo, and Y. Zhou, “**Over-the-air computation assisted hierarchical personalized federated learning**,” in Proc. IEEE International Conference on Communications (ICC), Rome, Italy, May 2023. [Paper]
- [2] F. Zhou, X. Chen, H. Shan, and Y. Zhou, “**Adaptive Transceiver Design for Wireless Hierarchical Federated Learning**,” in Proc. IEEE 98th Vehicular Technology Conference (VTC2023-Fall), Hong Kong, China, Oct. 2023. [Paper]
- [3] L. Wu, G. Gao, J. Yu, F. Zhou, Y. Yang, and T. Wang, “**PDD: Partitioning dag-topology dnns for streaming tasks**,” IEEE Internet of Things Journal, vol. 11, no. 6, pp. 9258–9266, Mar. 2024. [Paper]
- [4] F. Zhou, Z. Wang, H. Shan, L. Wu, and Y. Zhou, “**Over-the-Air Hierarchical Personalized Federated Learning**,” IEEE Transactions on Vehicular Technology, vol. 74, no. 3, pp. 5006–5021, Mar. 2025. [Paper]
- [5] F. Zhou, Z. Wang, Y. Shi, and Y. Zhou, “**Decentralized Satellite Federated Learning via Intra- and Inter-Orbit Communications**” in Proc. IEEE International Conference on Communications Workshops (ICC Wcshps), Denver, CO, USA, Jun. 2024. [Paper]
- [6] F. Zhou, Y. Shi, Y. Wu, S. Acharya, L. DaSilva, S. Kompella, W. Lou, and Y. T. Hou, “**WOS: An Optimized Scheduling Scheme for Federated Learning in Dynamic Wireless Networks**,” in Proc. IEEE Military Communications Conference (MILCOM), Los Angeles, CA, USA, Oct. 2025. [Paper]
- [7] F. Zhou, Y. Shi, W. Lou, and Y. T. Hou, “**FedHusky: Accelerating Hybrid Federated Learning with Client Hopping**,” in Proc. International Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks (WiOpt), Columbus, Ohio, USA, June 2026. [Paper]
- [8] F. Zhou, Y. Zhou, T. Wang, Y. Shi, C. Jiang, and H. Hu, “**Decentralized Satellite Federated Learning via Multi-Orbit Collaboration**,” submitted to IEEE Transactions on Wireless Communications. (Under Review)
- [9] F. Zhou, Y. Shi, Y. Wu, S. Acharya, L. DaSilva, S. Kompella, W. Lou, and Y. T. Hou, “**WOS+: Wait-for-Optimal-Set for Wireless Federated Learning**,” submitted to IEEE Journal on Selected Areas in Communications. (Under Review)
- [10] F. Zhou, Y. Shi, L. DaSilva, W. Lou, and Y. T. Hou, “**FedLAWN: Split Federated Learning in Low-Altitude Wireless Networks**,” submitted to IEEE Military Communications Conference (MILCOM), National Capital Region, USA, Oct. 2026. (Under Review)

## POSTERS

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- [1] F. Zhou, Y. Shi, Y. Wu, S. Acharya, L. DaSilva, S. Kompella, W. Lou, and Y. T. Hou, “**WOS: An Optimized Scheduling Scheme for Wireless Federated Learning**,” Poster presented at the 2026 CCI Symposium, Richmond, VA, USA, Apr. 2026. [Poster]
- [2] F. Zhou, Y. Shi, W. Lou, and Y. T. Hou, “**FedHusky: Accelerating Hybrid Federated Learning with Client Hopping**,” Poster presented at the 2026 ONR PI Meeting, St. Louis, MO, USA, May. 2026. [Poster]

## PATENTS

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[1] **F. Zhou**, Y. Shi, Y. T. Hou, “**A Method to Accelerate Federated Learning via Client Hopping,**” U.S. Provisional Patent Application filed. [[Link](#)]

## TEACHING ASSISTANT

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**ShanghaiTech University**

CS 287: Network Intelligence, Fall 2022

**Virginia Tech**

ECE 2714: Signals and Systems, Fall 2024 & Spring 2025

## AWARDS

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- COMAP’s Interdisciplinary Contest in Modeling (ICM), Honorable Mention (H Award), 2020
- CCI SWVA Cyber Innovation Scholarship, Commonwealth Cyber Initiative, Virginia Tech, 2026
- Student Travel Grant, IEEE/IFIP WiOpt, 2026

## SKILLS

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- **Programming**  
Python, Matlab
- **Languages**
  - Chinese (Native)
  - English (IELTS: 7.5; -Reading 7.5; -Listening 8.5; -Speaking 8.0; -Writing: 6.5)